Assignment #7: Final Report

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EDF5442: Inquiry and Measurement

Introduction

Training transfer is a critical component of clinical performance, patient outcomes, and organizational efficiency. However, training transfer is difficult to ensure due to several overlapping factors. Existing literature focuses on nursing staff and hospital environments, but other allied healthcare staff are under-studied. While some allied health professionals work in hospitals, others work in outpatient locations. Outpatient offices share these transfer priorities, but the types of pressures faced and environment in which the transfer takes place differs greatly between these sites, necessitating our focus on this special population as a subset of healthcare practitioners.

In this paper we present our exploration of the factors that influence training transfer among allied health professionals following a required, standardized, annual online job-related training. The project focused on three major influences identified in the literature; learner characteristics, training design, and workplace/supervisory environment, and utilized a mixed-methods approach to better understand how these factors support or hinder allied health professionals' ability to transfer newly learned skills to the workplace. It also considered relationships among and between the factors to understand how each one influences training transfer.

Literature Review

Literature Review

Training transfer, the application of newly learned skills to the workplace, has been widely studied across organizational learning research. Foundational models consistently identify three domains that shape transfer outcomes: learner characteristics, training design, and the workplace environment (Baldwin & Ford, 1988; Grossman & Salas, 2011; Holton, 1996). Although these predictors are well established, far less is known about how they operate within allied health professions, where clinical demands and workflow constraints create unique challenges. This review synthesizes prior research across the three domains and highlights gaps that motivate the present study.

Learner Characteristics

Learner characteristics such as motivation, readiness, prior knowledge, and self-efficacy strongly influence whether training is applied. Self-efficacy, defined as confidence in one's ability to perform a task, is one of the most consistent predictors of behavior change after training (Bandura, 1986). Holton (1996) similarly emphasizes motivation to transfer, noting

that even high-quality training fails when learners lack intention or confidence to use new skills. Healthcare research echoes these findings: clinicians are more likely to transfer training when they perceive it as relevant and aligned with their professional responsibilities.

However, most studies focus on nursing or general employee populations. Little is known about how occupational and physical therapists, whose work involves individualized treatment planning and rapid clinical decision-making, transfer newly acquired skills. Existing studies also tend to measure these constructs at a single time point, despite evidence that motivation and confidence shift throughout the training process. These gaps highlight the need to examine learner characteristics specifically within allied health contexts.

Training Design

Training design is another major determinant of transfer. Effective training includes clear objectives, realistic practice, and opportunities for feedback, all of which strengthen the likelihood that learners will use new skills on the job (Baldwin & Ford, 1988; Grossman & Salas, 2011). Meta-analytic evidence shows that training aligned with real job tasks produces significantly higher transfer rates than training that relies solely on passive formats such as lectures (Blume et al., 2010).

Despite this evidence, much of the healthcare literature focuses on simulation-based training or technical procedures, as in Tolsgaard et al (2016) and Duarte et al (2023). Less attention is paid to the mandatory, hospital-wide modules that allied health professionals complete annually. These modules often rely on slide decks or computer-based activities that may not provide the practice or feedback opportunities known to support transfer. While Gregory et al (2022) conducted their research in a hospital environment to determine how training design impacted transfer, the study design focused on interviews with key informants instead of a comprehensive view. Few studies examine how allied health clinicians perceive the realism or applicability of these formats, leaving a gap in understanding how training design influences transfer in everyday clinical settings.

Workplace and Supervisory Environment

The transfer climate, including supervisor support, peer encouragement, workload, and resource availability, plays a critical role in determining whether training is used (Grossman & Salas, 2011). Supervisors follow-up, in particular, has been shown to increase the likelihood that employees will attempt new behaviors (Park et al., 2018). In healthcare

settings, transfer climate is especially influential due to staffing constraints, documentation demands, and unpredictable patient needs.

Yet research on allied health environments remains limited. Most studies examine hospital-wide organizational factors or nursing units, with little attention to outpatient rehabilitation, acute care therapy departments, or interdisciplinary therapy teams. Understanding how clinicians navigate productivity expectations, equipment availability, and scheduling pressures is essential for identifying real-world barriers to transfer.

Gaps and Rationale for the Present Study

Across the literature, three gaps stand out. First, few studies examine all three domains simultaneously, even though transfer is multidimensional. Second, allied health professionals, particularly OT and PT clinicians, are underrepresented in transfer research. Third, most studies rely solely on quantitative methods, limiting insight into the contextual factors that shape clinicians' decisions to apply training.

The present study addresses these gaps by using a mixed-methods approach to examine how learner characteristics, training design, and workplace conditions influence training transfer among allied health professionals who recently completed job-related training. This approach provides a more comprehensive understanding of the supports and barriers that shape transfer in real clinical environments.

Research Questions

To better understand the training experience and perception of transfer for allied health professionals, the following over-arching research question was established, along with a subset of questions that allowed for comparison between each of the three factors, using Baldwin and Ford (1988) as a framework for our research.

Primary RQ: What factors influence training transfer among allied health professionals who have recently completed job related training?

> RQ1.1 - What impacts do learner characteristics have on transfer among allied health professionals?
>
> RQ1.2 - What impact does training design have on transfer among allied health professionals?
>
> RQ 1.3 - What impact does the workplace and supervisory environment have on transfer among allied health professionals?

The first research question allows us to examine the learner's motivation, confidence, readiness, and background knowledge in relation to their perception of transfer to the workplace. The second research question isolates the impacts of the training experience itself, the presence of clear objectives, opportunities for practice and feedback, and a realistic context that allows for near and far transfer of knowledge, or generalization. Finally, the third research question focuses on the overall environment in which the transfer would take place, including whether the peer and supervisory environment support training objectives and provided the resources to apply and maintain those skills after the training has ended.

Methods

Participants and Recruitment

Our recruitment approach began with a practical connection: Erin's sister is an occupational therapist, and her clinic regularly completes the same annual hospital-wide training that inspired our research interest in training transfer among allied health professionals. Because the entire clinic staff, therapists, assistants, front desk staff, and managers, are required to complete this training each year, we expected this setting to provide a strong pool of potential participants. Erin's sister agreed to share the Qualtrics survey link with her close colleagues, and her supervisor agreed to distribute the link to the full clinic staff.

Despite these efforts, our final survey sample was much smaller than anticipated. A total of seven healthcare professionals completed the survey. While this number was lower than we hoped for, it still provided enough data to examine patterns related to our research questions within the scope of a small inquiry project.

For the interview portion, our goal was to gather deeper, experience-based insights from clinicians who had recently completed job-related training. Each team member selected one clinician from the pool given by Erin's sister, Allison. Three clinicians participated: one occupational therapist and two physical therapists, all with more than 20 years of clinical experience. Because of their demanding schedules, interviews were conducted asynchronously through email, which allowed participants to respond at their convenience and provide more thoughtful, detailed answers.

Survey Design and Development

We created the survey in Qualtrics and designed it around the three major areas identified in the training transfer literature: learner characteristics, training design, and the workplace environment. To make sure the items aligned with established research, we drew directly from the Baldwin and Ford Transfer of Training Model, Holton's HRD Evaluation Model, and Social Cognitive Theory. These frameworks helped us decide which constructs to measure and how to phrase the items.

The final survey included:

- 5 demographic questions
- 3 multiple-choice questions about the training itself
- 3 multiple-choice questions about transfer-related conditions
- 12 Likert-scale items (1–5) aligned with our three research questions
- 2 open-response questions asking participants to describe training activities and examples of applying (or not applying) what they learned

The survey took about 5 minutes to complete. The survey remained open throughout our recruitment period, and analysis began once responses stabilized. A copy of the full survey instrument is available in Appendix A.

Survey Data Analysis

Quantitative data were analyzed using descriptive statistics provided by Qualtrics, including means, standard deviations, and response frequencies. These statistics helped us identify patterns related to each research question.

For the open-response items, we used content analysis. Each team member reviewed the responses and noted recurring ideas. We then compared our notes and grouped similar ideas into broader themes (such as relevance, practice opportunities, or supervisor support). These themes were later used to support and expand on the quantitative findings.

Interview Design and Procedures

The interview protocol was designed to complement the survey by giving us more detailed examples of how clinicians experience training transfer. The questions were organized around the same three research questions and included prompts about readiness, motivation, training design features, and workplace conditions.

All interviews were conducted asynchronously by email, which allowed participants to respond when they had time and to give more thoughtful answers. Each participant

received an opening script, consent information, and the full set of interview questions in one message. We followed up with clarification questions when needed. All three interviewees responded to every question. A copy of the full interview instrument is available in Appendix B.

Interview Data Analysis

We analyzed the interview data using thematic analysis. Each team member coded their own interview first, identifying ideas related to learner characteristics, training design, and workplace factors. We then met to compare codes and agree on a shared set of themes. This process helped us make sure we were interpreting the responses consistently.

After finalizing the themes, we organized them by research question so they could be compared directly with the survey findings. This allowed us to use the interviews to explain or expand on patterns we saw in the quantitative data.

Alignment With Research Questions

Both the survey and interview were intentionally designed to align with our three research questions:

- **RQ1.1 (Learner Characteristics):** Items on readiness, motivation, confidence, and background knowledge; interview questions about preparedness, motivation, and relevance.
- **RQ1.2 (Training Design):** Items on clarity of objectives, practice opportunities, feedback, and envisioning skill use; interview questions about realism, practice, and design features.
- **RQ1.3 (Workplace Environment):** Items on supervisor support, peer support, resources, and changes in practice; interview questions about workload, culture, and supervisory influence.

This alignment ensured that both data sources contributed directly to answering each research question.

Results

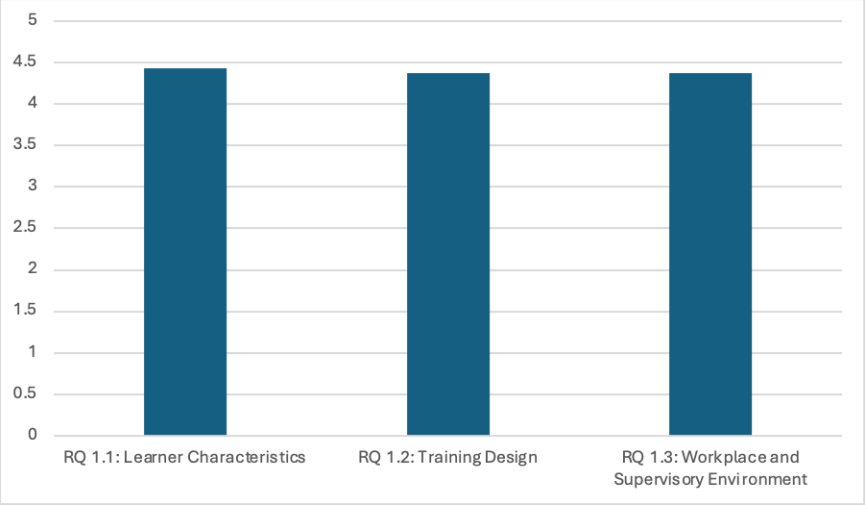
Survey Findings

The survey was completed by 7 participants in allied health professions. Participants were primarily physical therapists and physical therapy assistants (57%, n=4) or occupational therapists and occupational therapy assistants (43%, n=3), based on participants' self-described roles, in clinical outpatient settings, rehabilitation, and acute care/hospital. Respondents are either employed full-time (86%, n=6) or part-time (14%, n=1) with a wide range of experience in the field and at the specific work site. 71% (n=5) had been employed in their healthcare profession for more than 15 years, with 14.5% (n=1) reporting 0-2 years of experience and 14.5 (n=1) reporting 3-5 years of experience. 43% (n=3) reported more than 11 years at the same facility, 29% (n=2) reported 1-3 years at the facility, and 14% each reported less than 1 year and between 4-6 years of experience at their current workplace. Because some demographic items were open-ended, clinical settings were categorized based on participants' written descriptions.

In general, there was a high level of agreement for each factor studied in this project. Aggregated means exceeded 4.3 out of 5, indicating participants agreed that each factor had a high impact on their training. This supports Baldwin and Ford's (1988) conception of three primary factors that influence training transfer holds true for allied health professionals.

Figure 1

Aggregated means by factor of training transfer



Due to the limited sample size for this project, we could not confirm whether any one of these factors was significantly more influential than the other two for the allied health population. Table 1 below summarizes each Likert item present in the survey, the specific feature of training transfer, its associated factor, and its individual mean score. These items

were sequenced chronologically to more closely mirror the learner's order of experience, instead of being organized by our research questions, to help bring participants back through their experience methodically. Overall, the highest-rated factor was clear training outcomes (4.9), a training design factor, with the next highest factors being clinician confidence and background knowledge (4.7), which are learner characteristics. The lowest-rated factors were opportunities to practice (3.9) in the training design, the learner characteristic eagerness to learn (3.9), and overall behavioral change (4.1), a workplace factor.

Table 1

Likert items and means by transfer factors and specific features.

Specific Feature	Factor (RQ)	Mean Score (of 5)
Clear training outcomes	Training Design (RQ 1.2)	4.9
Opportunities to practice	Training Design (RQ 1.2)	3.9
Constructive feedback	Training Design (RQ 1.2)	4.0
Skill generalization	Training Design (RQ 1.2)	4.7
Eager to learn	Learner Characteristics (RQ 1.1)	3.9
Opportunity to apply	Learner Characteristics (RQ 1.1)	4.4
Confident in skills	Learner Characteristics (RQ 1.1)	4.7
Background knowledge	Learner Characteristics (RQ 1.1)	4.7
Supervisor encouragement	Workplace and Supervisory (RQ 1.3)	4.4
Colleague support	Workplace and Supervisory (RQ 1.3)	4.6
Necessary resources	Workplace and Supervisory (RQ 1.3)	4.4
Behavior change	Workplace and Supervisory (RQ 1.3)	4.1

RQ 1.1: What impacts do learner characteristics have on transfer among allied health professionals?

Within this factor, we discovered that learners felt confident in their ability to use the skills they learned (4.7), had necessary background knowledge (4.7), and were eager to apply what they had learned (4.7). However, the learner's self-assessed eagerness to learn was the lowest-scoring item in this research question (3.9). The training we isolated for study was a required annual training module for healthcare practitioners, potentially limiting intrinsic interest in the material, and since most of the clinicians had substantial experience in the field and many years of experience at their current workplace, their general background knowledge and confidence may have been higher due to prior exposure to this annual training material, despite the presence of new or updated learning content.

Through post-hoc analysis, we found a positive correlation between learner confidence and interest in opportunities to apply knowledge learned (p=.0285, effect size .806), and a separate positive correlation between existing background knowledge and interest in opportunities to apply knowledge learned (p=.0285, effect size .806). Figures 2 and 3 demonstrate the relationships between these variables; however, due to the limited number of respondents for the project, these findings can only be considered representative of the specific sample. Further research with a larger allied health audience would provide an opportunity to determine whether this trend holds and can be generalized to allied health practitioners more broadly; qualitative evidence to support this connection would be valuable.

Figure 2

Positive correlation between opportunities to apply knowledge and background knowledge

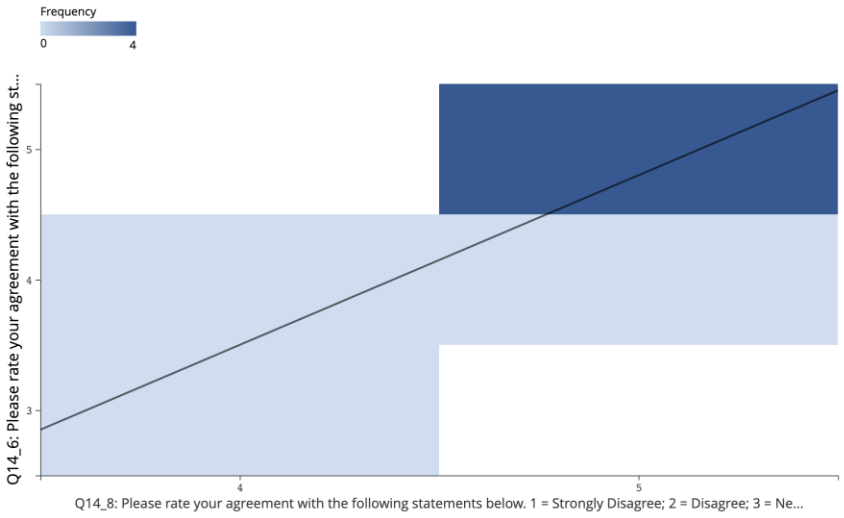
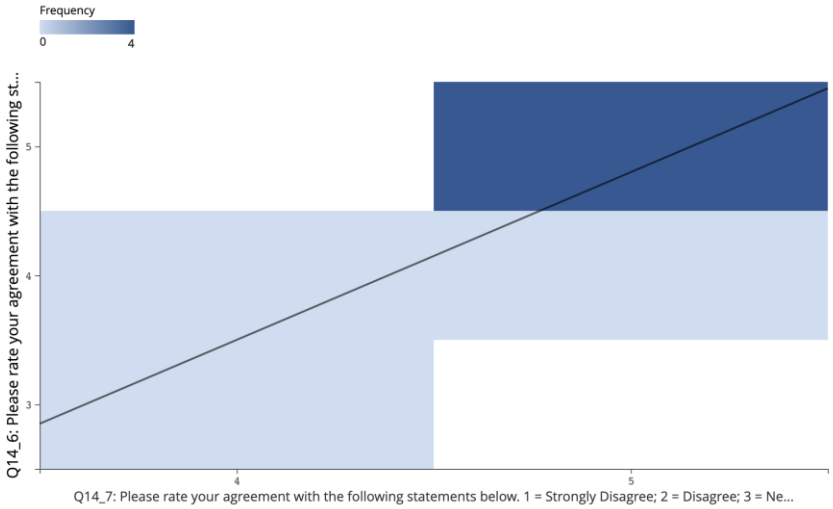


Figure 3

Positive correlation between opportunities to apply knowledge and learner confidence



In this project, we did not find any significant relationships between the items associated with learner characteristics and overall perceived behavioral change. If a larger sample becomes available for future research, this finding could be re-evaluated.

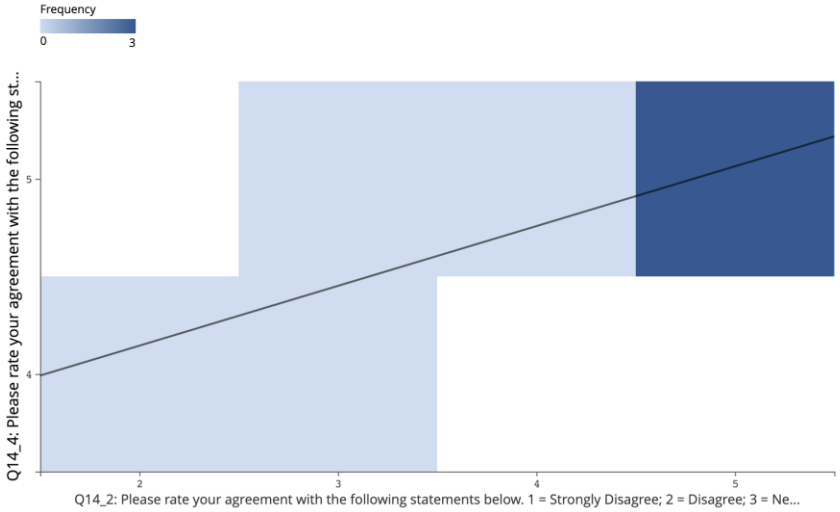
RQ 1.2: What impact does training design have on transfer among allied health professionals?

Within this factor, we found that learners felt that certain training design outcomes were more present than others. The participants agreed that the training included clear outcomes (4.9) and provided examples that allowed them to generalize the skills to the workplace (4.7). However, opportunities to practice and the presence of constructive feedback were lacking, possibly due to the nature of the self-paced annual online module, which may have had limited opportunity for interactive practice despite being well organized, engaging, and representative of the workplace.

As with the prior research question, a significant relationship was found within two items related to this factor. There was a positive correlation between opportunities to practice and the ability to envision (or generalize) use of the knowledge or skill in the workplace ($p=.046$, effect size .763). Figure 4 below illustrates this relationship between variables.

Figure 4

Positive correlation between opportunities to practice and skill generalizability to workplace



The small sample size limits the overall generalizability of this observation. It is a trend that would benefit from additional investigation. No significant relationship was found between the training design factors and overall perceived behavioral change.

RQ 1.3: What impact does the workplace and supervisory environment have on transfer among allied health professionals?

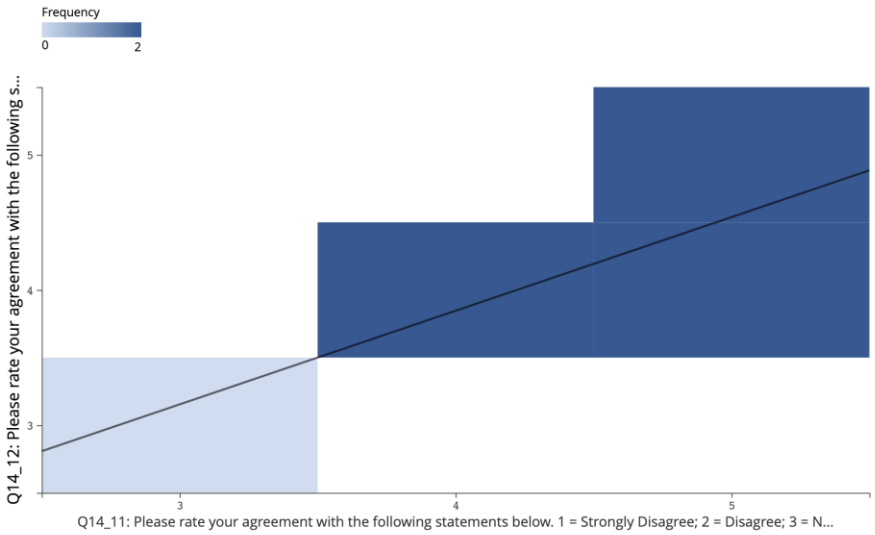
When examining the final factor, we discovered that colleague support (4.6) was perceived as a higher influence on training transfer than supervisor support (4.4) or available resources (4.4). Interestingly, respondents rated the statement focused on behavioral change as a result of training as the lowest item in this factor. Again, the focus of the required training module on procedures such as hand washing, HIPAA updates, intercom codes, and emergency procedures for lockdown situations may have influenced these responses. Some of the content presented is simply required (handwashing, HIPAA compliance) whereas others are only needed situationally (codes, lockdowns); this may have influenced how participants responded about their behavioral changes due to training specifically. In general, this factor had the lowest range in means, which may indicate that within this sample, the supervisory and workplace environment was a stabilizing force in training transfer.

However, this research question contained the only significant relationship between behavioral change and another item as a significant driver of change. The presence of available resources to implement what was learned in training was positively correlated with overall perceived behavioral change (p=.0348, effect size .789). Interestingly, this does

not align with the respondents' perception of their colleagues as the most influential element within this factor, and this mismatch deserves further investigation in future.

Figure 4

Positive correlation between resource availability and perceived behavioral change



As with the other research questions, a follow up study with a larger sample would help to confirm these relationships.

Additional investigation into the role of part-time versus full-time employment could potentially bear fruit. Only one part-time staff member completed the survey, but the respondent had statistically significant differences compared to full-time staff on items related to supervisory support and opportunities to apply their new knowledge. There were also two items where the part-time employee's responses were far lower than the mean for the full-time group – in whether the training design provided opportunity for practice and clear feedback, and in their motivation to complete the training. The part-time employee's responses may have skewed the relationships between specific features of learner characteristics or training design. These responses were intentionally kept due to the sample size, but further investigation is warranted on the generalizability to part-time allied health professionals or the degree of influence of this potential outlier on our results.

In the open-response portion of the survey, participants highlighted themes related to training design, emphasizing the overall generalization of the learning content and the value of its application; one participant highlighted the relevance of the learning to their daily practice, noting *"I use it every day in my work…when a code is called, I feel confident knowing what that code means."* A secondary theme centered around the importance of

the training activities that influence behavior, specifically reminders for handwashing procedures. With a larger sample, additional thematic analysis may highlight additional themes we did not originally encounter.

Interview Findings

RQ 1.1: Learner Characteristics

Across interviews, clinicians consistently described high readiness and confidence prior to completing their required or elective training. One clinician explained that they felt "a high level of readiness" because they had "been a PT for 26 years" and were accustomed to completing similar annual modules. Another clinician described feeling ready because they "do a module every year" and had "worked for this hospital for almost 28 years," noting that much of the content was familiar.

Motivation to apply the training varied depending on whether the material introduced new skills. One clinician shared that they "really didn't have to change what I did, as I had already been doing what was instructed," though the training served as "a good reminder that what I'm doing is important." Another clinician, who attended a specialty orthopedic update, reported being "motivated to try out certain new techniques with patients" and described being "excited to learn more" after reviewing the agenda.

Confidence was consistently high across participants. One clinician stated they felt "highly confident," noting that repeated annual exposure increased their comfort over time. Another clinician described how confidence "increased over time" as they applied new orthopedic techniques and observed patient responses. Overall, clinicians emphasized that relevance, experience, and repetition shaped their motivation and confidence.

RQ 1.2: Training Design

Clinicians described clear differences between the design of the annual online module and the specialty orthopedic training. For the online module, one clinician highlighted that the training was "clear & concise," with "cute animation" and short scenario-based examples that "helped to keep me engaged." They also noted that the end-of-section questions reinforced learning: "I was able to easily get them correct as the training was engaging."

Another clinician appreciated that the most recent module included "more interactive" testing elements, explaining that "you didn't just select the answer… there were things you had to do using the mouse." However, this same clinician noted that the hour-long format led to reduced engagement toward the end: "I was getting really bored and thus don't feel I was as engaged toward the end as I was at the start."

In contrast, the orthopedic update was described as highly relevant and directly applicable. One clinician explained that the material was “easily transferable to the clinic” because the examples and treatment options aligned closely with their outpatient caseload. They emphasized that the organization of the material and the practical examples “made it easier to remember or continue using the skills.”

Across interviews, clinicians highlighted that design features such as realistic examples, interactive elements, and clear organization supported transfer, while long duration and lack of hands-on practice reduced engagement.

RQ 1.3: Workplace and Supervisory Environment

Clinicians described workplace environments that generally supported the application of training. One clinician noted that working with a team “collectively skilled and committed to high quality patient care” helped reinforce correct procedures, explaining that “seeing the other therapists doing the correct procedures consistently helps it to become a habit.”

Supervisory support was also described positively. One clinician shared that their supervisor “offers his support if we have any challenges and offers reminders when needed,” including facilitating access to needed equipment. Another clinician described a supervisor who “expects for us as clinicians to be competent & keep up with the latest information,” and who regularly checked in to ensure staff had time to complete required training.

Workload and time pressures were identified as potential barriers. One clinician explained that “less staff/higher caseloads/more paperwork tends to make staff rush more and mistakes can be made or shortcuts taken.” Another clinician noted that being “very busy and short staffed” made it harder to try new techniques, though they sometimes adjusted treatment sessions or home programs to create space for new approaches.

Workplace culture was consistently described as supportive. One clinician stated that colleagues “support it & we will also remind each other” when applying new procedures. Another clinician described a culture where staff “enjoy learning from one another” and often share new techniques through brief in-services.

Summary of Qualitative Themes

Across interviews, three themes emerged:

- **Experience-driven readiness and confidence**: clinicians with long tenure felt prepared and confident, especially for recurring annual modules.

- **Design influences engagement**: interactive elements, clear organization, and relevant examples supported transfer, while long or repetitive modules reduced engagement.
- **Supportive environments enable transfer**: peer reinforcement, supervisory encouragement, and access to resources facilitated application of training, though workload pressures sometimes limited opportunities to try new skills.

These findings provide a detailed qualitative complement to the survey results, illustrating how allied health professionals experience training transfer within real clinical settings.

Discussion

This mixed-methods study examined how learner characteristics, training design, and the workplace/supervisory environment support training transfer among allied health professionals completing required job-related training. Overall, clinicians reported positive perceptions in all three domains, with especially strong confidence and background knowledge, clear training objectives, and supportive colleagues and supervisors, alongside comparatively weaker ratings for practice opportunities, feedback, and self-reported behavior change.

Across all 12 Likert items, mean scores exceeded 4.1 on a 5-point scale, indicating generally positive perceptions of the training, learner readiness, and the environment for applying new skills. Clear objectives, strong self-efficacy and background knowledge, and supportive colleagues and supervisors emerged as consistent strengths, whereas opportunities for practice, constructive feedback, and reported behavior change were relatively weaker points. Exploratory correlations showed that higher confidence and background knowledge were associated with stronger intentions to apply training; more practice was associated with greater ability to envision using skills at work; and greater resource availability was associated with higher self-reported behavior change. Interview data reinforced these patterns, highlighting experience-driven readiness and confidence, design features that shape engagement, and workplace culture and resources as key influences on transfer.

Learner characteristics and transfer (RQ1.1)

Both survey and interview findings underscore the importance of learner characteristics in supporting training transfer among allied health professionals. Survey respondents reported very high confidence in their ability to use skills learned (M = 4.7) and strong background knowledge (M = 4.7), while intention to look for opportunities to apply learning

was also high (M = 4.4). Clinicians in this study entered training with high levels of readiness and self-efficacy, and these characteristics were closely tied to their intentions to apply what they learned. Even though motivation to complete the required annual module was more modest, especially for the part-time clinician, participants still expressed strong intentions to look for opportunities to apply key content. This pattern aligns with Social Cognitive Theory and prior transfer research, which identify self-efficacy and perceived relevance as central drivers of behavior change (Bandura, 1986; Holton, 1996; Blume et al., 2010).

Clinicians' interview comments reinforce this interpretation; many described "high readiness" largely because they had completed similar modules for years and saw them as part of routine professional expectations. When training introduced genuinely new clinical techniques, such as in the orthopedic update, motivation to apply learning increased further, suggesting that novelty plus relevance can amplify the transfer effect for already confident practitioners. Together, these findings suggest that for experienced allied health professionals, confidence and perceived utility may matter more than intrinsic enthusiasm for a mandatory module in predicting whether they will use what they learned.

A notable and somewhat unexpected pattern emerged in the contrast between high self-efficacy and only moderate eagerness to learn. Despite lower enthusiasm for a mandated module, clinicians still reported strong intentions to apply content, suggesting that for experienced allied health professionals, perceived relevance and confidence may be more important than intrinsic interest in driving transfer. This aligns with findings from Gegenfurtner (2011) and Park et al. (2018), who indicate that motivation is multifaceted and that perceived utility and support can sustain transfer-related behaviors even when initial enthusiasm is modest.

Training design and transfer (RQ1.2)

Training design showed a combination of strong and weaker elements. The annual online module was perceived as clear, concise, and relevant, with very strong agreement that objectives were explicit. The highest-rated feature in the entire survey was clear training outcomes (M = 4.9), and clinicians also strongly agreed that they could envision using the skills in their workplace (M = 4.7). These results align with Grossman and Salas, and Baldwin and Ford, who emphasize clear objectives and realistic, job-relevant content as core design elements that support transfer.

At the same time, clinicians reported gaps in practice and feedback opportunities, which likely reflect the constraints of a self-paced online module. Respondents rated opportunities to practice new skills (M = 3.9) and clear, constructive feedback (M = 4.0)

lower than other design elements. Interview accounts help explain this pattern, while the required online module was perceived as “clear & concise” and engaging, with animations and interactive quiz questions, clinicians also reported becoming “really bored” toward the end of the hour-long module and noted limited hands-on or scenario-based practice compared to more immersive trainings. In contrast, a specialty orthopedic update was described as highly relevant, organized around real clinical cases, and easily transferable to day-to-day practice, reinforcing the value of design features that closely mirror clinical tasks.

Both survey data and interviews suggest that when training includes richer practice and interactive elements, it becomes easier for clinicians to imagine and plan for real life use. This is consistent with research on simulation-based and digital training, where realistic practice, learner control, and meaningful feedback are associated with stronger self-efficacy and transfer (Gegenfurtner et al., 2014; Tolsgaard et al., 2016). In this study, the contrast between the annual module and the orthopedic update further illustrates that design features that closely mirror clinical cases and workflows are especially powerful in supporting transfer.

At the same time, the absence of a significant relationship between training design items and overall behavior change in this small sample suggests that design alone may not be sufficient to drive observable changes, particularly when content reiterates existing procedures (e.g. handwashing, HIPAA compliance, emergency codes) rather than introducing novel skills.

Workplace and supervisory environment and transfer (RQ1.3)

The workplace and supervisory environment were generally perceived as supportive, with clinicians describing strong peer reinforcement, constructive supervisory expectations, and reasonable access to resources. These conditions align with the concept of transfer climate and social support outlined by Grossman and Salas (2011), Chiaburu et al. (2010), and Rouiller and Goldstein (1993), where colleagues, supervisors, and organizational systems signal whether using new skills is expected and valued.

Workplace and supervisory factors were perceived as generally supportive, but with important nuances. Survey respondents rated colleague support (M = 4.6), supervisor encouragement (M = 4.4), and availability of resources (M = 4.4) relatively high, consistent with prior research on transfer climate and social support (Grossman & Salas, 2011; Chiaburu et al., 2010; Rouiller & Goldstein, 1993). The item assessing behavior change (“I can identify specific changes I have made to my evaluation or treatment plans because of training”) received a lower mean of 4.1, which is still positive but comparatively modest.

The most notable quantitative finding in this domain was the positive correlation between resource availability and perceived behavior change ($p = .0348$, effect size = .789). This suggests that clinicians who perceive sufficient equipment, materials, and staffing/scheduling support are more likely to report concrete changes in their evaluation or treatment plans after training. Qualitative data further support this relationship, interviewees described supervisors who facilitated access to needed equipment and colleagues who reinforced correct procedures, emphasizing that practical supports made it easier to apply and sustain new behaviors.

Within this overall positive climate, two nuances stand out. First, available resources (such as equipment, materials, and scheduling flexibility) were closely tied to whether clinicians reported changing their evaluation or treatment practices. This suggests that tangible supports may be the immediate bridge between intention and behavior, even when social support is strong. Second, workload and contextual fit occasionally constrained transfer; clinicians highlighted that high caseloads and documentation demands limited opportunities to try new approaches, and outpatient clinicians noted fewer opportunities to apply hospital-based content. These findings echo prior work showing that supportive climates must be coupled with realistic workload and context-appropriate content to fully realize transfer (Clarke, 2002; Gregory et al., 2022).

Interestingly, the connection between resources and behavior does not fully align with respondents' perception that colleagues were the most influential element within this factor, indicating a potential discrepancy between what clinicians consciously attribute influence to (social support) and what quantitatively predicts reported behavior change (tangible resources). This mismatch may reflect the more subtle effect of resources as less immediately noticeable. Clinicians may take equipment and staffing levels for granted until they become constraints. It also resonates with qualitative reports that workload, staffing shortages, and documentation demands can make it difficult to try new techniques, even in otherwise supportive environments.

Finally, taken together, the findings support the multidimensional view of training transfer proposed by Baldwin and Ford and Holton; learner characteristics, training design, and the work environment operate together rather than independently. In the study, learner readiness and self-efficacy were strong, training objectives and relevance were clear, and the workplace climate was generally supportive, yet reported behavior change, while positive, was not as strong as these favorable conditions might predict.

One plausible explanation is that the type of training matters. The required annual module largely reinforced established procedures (hand hygiene, HIPAA, emergency codes), which may have limited the scope for observable behavior change among already experienced

clinicians. In contrast, when trainings introduced new, practice-changing content (such as specialized orthopedic techniques) clinicians described higher motivation and more concrete changes in their day-to-day work. This suggests that alignment among learner readiness, novel yet relevant content, practice-oriented design, and adequate resources may be the key combination that produces the strongest transfer in allied health settings.

An exploratory pattern involving the single part-time clinician also points to potential inequities. This respondent reported lower motivation, less favorable perceptions of practice and feedback, and weaker supervisor support than full-time peers. While no conclusions can be drawn from one case, this hints that employment status and scheduling may shape exposure to, and support for, training in ways that warrant further study.

Instructional Implications

The findings suggest several practical steps for improving training transfer for allied health professionals.

1. Build on strong self-efficacy and experience: Experienced clinicians already feel confident and prepared; training should explicitly connect new content to their existing expertise and clinical judgment. Brief prompts that link modules to typical cases, and opportunities for clinicians to reflect on how content fits into their current practice, can further reinforce readiness to transfer.

2. Enhance active practice and feedback in online modules: Required modules should go beyond passive viewing by adding realistic, scenario-based practice and clearer feedback on performance. For example, branching case vignettes, interactive code simulations, and feedback that explains how choices affect patient outcomes can make learning more actionable, and presenting content in shorter, modular segments can help maintain engagement and reduce fatigue.

3. Align training content with specific settings and roles: Training on hospital-wide topics should include clear examples for both inpatient and outpatient contexts so that all clinicians can see how to apply concepts in their own workflow. Where feasible, tailored versions or supplemental modules for specific settings (e.g. outpatient rehabilitation) can reduce the gap between generic content and local practice.

4. Formalize supervisor follow-up and peer reinforcement: Even in supportive climates, transfer can be strengthened by making follow-up more intentional. Brief

supervisor-led debriefs, case discussions tied to recent training, and structured opportunities for staff to share how they are applying new skills can keep training content visible amid competing demands.

5. Pair training with resource and workload planning: Resources are closely linked to reported behavior change, therefore training initiatives should include checks on equipment availability, charting tools, and scheduling flexibility. When constraints cannot be fully resolved, training should acknowledge these instances and offer concrete strategies for adapting within limitations.

6. Attend to part-time and variably scheduled staff: Organizations should examine whether part-time clinicians receive equivalent access to training, practice opportunities, and supervisor feedback. Flexible scheduling for modules and deliberate inclusion in follow-up activities may help ensure that transfer support reaches all members of the team.

Overall, this study suggests that allied health training transfer is most strongly supported when confident, experienced clinicians receive clear, context-relevant training that includes opportunities for practice and feedback, and when their workplaces provide both social support and the concrete resources needed to use new skills in demanding clinical environments.

Limitations

Like most small-scale inquiry projects, this study has several limitations that should be considered when interpreting the findings. The most significant limitation is the small sample size. Although we attempted to recruit a larger group of participants through an allied health clinic, only seven individuals completed the survey. This limited the range of perspectives represented and reduced our ability to examine differences across roles, settings, or experience levels. The three interview participants provided rich detail, but their experiences may not reflect the broader allied health population.

Another important limitation involves the interpretation of the quantitative findings. With only seven survey respondents, the descriptive statistics are useful for identifying general patterns, but they should not be interpreted as strong evidence of differences between groups or as reliable indicators of relationships between variables. Although we reported means and noted patterns across items, the sample size was too small to draw firm conclusions about subgroup differences or to treat any observed trends as representative of allied health professionals more broadly. This aligns with the feedback from our previous

assignment and reinforces the need for caution when interpreting correlations or apparent group patterns in such a limited dataset.

A second limitation relates to sampling and recruitment. Because we relied on convenience sampling through personal and professional networks, the participants who chose to respond may differ from those who did not. For example, clinicians who are more comfortable with training or more engaged in their workplace may have been more likely to participate. This introduces the possibility of self-selection bias.

The survey instrument also presents limitations. Although the items were grounded in established training transfer frameworks, the survey was brief and designed to be completed quickly by busy clinicians. As a result, some constructs, such as motivation to transfer or supervisor support, were measured with only one or two items. This limits the depth and nuance of the quantitative data. In addition, because the survey asked participants to reflect on a recent required training, responses may have been influenced by recall bias or by the nature of mandatory modules, which often differ from more interactive or clinically focused trainings.

The qualitative data have limitations as well. Interviews were conducted asynchronously through email, which allowed participants flexibility but sometimes resulted in shorter or less elaborated responses than a live interview might have produced. Our analysis process was collaborative and discussion-based rather than a formal multi-coder approach, which is appropriate for a class project but may reduce the reliability of theme identification.

Finally, the study's context limits generalizability. All participants worked in healthcare settings within a single region, and many were connected through the same clinic network. Training transfer is shaped by organizational culture, staffing patterns, and workflow demands, all of which vary widely across facilities. As a result, the findings should be interpreted as exploratory rather than representative of all allied health environments.

Despite these limitations, the combination of survey and interview data provides meaningful insight into how allied health professionals experience training transfer and highlights areas for future research with larger and more diverse samples. Further study with a larger sample would strengthen these conclusions and add to the existing literature on training transfer for allied health professionals.

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Appendix A: Survey Instrument

Q1 Welcome and thank you for participating! This survey asks about your most recent hospital-wide required training and how you have applied it in your clinical work. Please answer all questions based on that specific training experience. Your responses are confidential and the survey takes about 5 minutes to complete.

Q2 What is your primary allied health profession?

_____

Q3 What type of clinical setting do you work in?

_____

Q4 What is your employment status?

- o Full Time
- o Part Time
- o PRN/Per Diem

Q5 How many years have you worked in your profession?

- o 0-2 years
- o 3-5 years
- o 6-10 years
- o 11-14 years
- o 15+ years

Q6 How long have you worked at your current facility?

- o Less than 1 year
- o 1-3 years
- o 4-6 years
- o 7-10 years
- o 11+ years

Q7 Please continue to think about the most recent hospital-wide training required for all staff.

Q8 When did your most recent training occur?

- o Within the past month
- o 1-3 months ago
- o 4-6 months ago
- o 7-12 months ago

Q9 How was the training delivered?

- o In-person
- o Online (remote meeting such as Zoom)
- o Online (another format, such as an online course)
- o Other

Q10 What type of training was it?

- o Open-skills: learning "soft" skills (team dynamics, leadership, communicating with patients and families)
- o Closed-skills – learning a step-by-step process (ex. EPIC training, OSHA standards, infection control, measuring vital signs)
- o Both Open and Closed Skills
- o Other (specify) _____
- o Unsure

Q11 Before starting the training, how confident were you that you could successfully apply what you learned?

- o Not confident
- o Slightly Confident
- o Moderately Confident
- o Very Confident

Q12 After you complete job-related training, how often does your immediate supervisor follow up with you (to discuss how you will apply the training, review cases, or provide feedback)?

- o Never
- o Rarely
- o About half the time
- o Most of the time

- o Always

Q13 To what extent does your organization provide the time and resources you need to apply new skills from training in your clinical work?

- o Not at all
- o To some extent
- o To a great extent
- o Completely

Q14 Please rate your agreement with the following statements below.
1 = Strongly Disagree; 2 = Disagree; 3 = Neutral; 4 = Agree; 5 = Strongly Agree

The training module clearly stated what I should be able to do after completing the training.	☆ ☆ ☆ ☆ ☆
The training included sufficient opportunities to practice new skills.	☆ ☆ ☆ ☆ ☆
I received clear and constructive feedback during training activities.	☆ ☆ ☆ ☆ ☆
After the training, I could envision myself using the skill in the workplace.	☆ ☆ ☆ ☆ ☆
I was eager to learn the material presented in this training.	☆ ☆ ☆ ☆ ☆

I plan to look for opportunities to apply what I learned.	☆	☆	☆	☆	☆
I am confident in my ability to use the skills learned in this training.	☆	☆	☆	☆	☆
I had the necessary background knowledge to benefit from this training.	☆	☆	☆	☆	☆
My clinical supervisor encourages me to use the skills and techniques I learn in training with my patients.	☆	☆	☆	☆	☆
Colleagues in my unit generally support trying out new evidence-based techniques learned in training.	☆	☆	☆	☆	☆
I have the necessary resources (including sufficient equipment, materials, and scheduling/staffing	☆	☆	☆	☆	☆

support) to implement and refine skills learned in the training.	
I can identify specific changes I have made to my evaluation or treatment plans because of training.	☆ ☆ ☆ ☆ ☆

Q15 Please describe the training activities you encountered. Common activities may include (but not be limited to) slide presentations, video-based skills demonstrations, role-play, computer-based simulation exercises, and more.

_____

_____

_____

_____

_____

Q16 Can you describe a situation where you used, or chose not to use, something from this training? What influenced that decision?

_____

_____

_____

_____

_____

Appendix B: Interview Questions

Opening Script

Hello, and thank you for agreeing to participate in this interview. Our team is conducting a small inquiry project for a graduate course at Florida State University. We are exploring how allied health professionals experience job-related training and what factors influence whether new skills are applied in practice.

For this interview, please think specifically about the recent hospital-wide training you completed. All responses should be based on that training experience.

Your participation is completely voluntary, and your responses will remain confidential. Your name or identifying details will not appear in any report. This interview should take about 10 minutes to complete. Please answer the questions in as much detail as you feel comfortable sharing and return your responses by email no later than March 29, 2026.

If you have questions at any point, feel free to let me know. When you're ready, you may begin responding to the questions below. Thank you again for your time!

Interview Questions

Note on Probes

The following interview protocol includes possible probes listed under each question. These probes reflect how the interviewer would seek clarification or additional detail if needed. Because this interview is conducted asynchronously via email, probes will only be sent in a follow-up message if a response requires clarification.

Opening Question:

1. To start us off, can you please briefly describe the most recent training you attended, the topic, and what relevance it has for your role and workplace? Have you used the skills you learned? If so, how? And if not, why?

   Probe:

   - Can you share a specific example of how you used (or didn't use) the training?

Learner Characteristics:

2. Before completing the training, how prepared or "ready" did you feel to learn the material? What contributed to that feeling?

   Probes:

   - What influenced that level of readiness?
   - Was there anything that helped or hindered you?

3. How motivated were you to apply the skills from the training once you returned to your clinical work? What increased or decreased that motivation?

   Probes:

   - What affected your motivation the most?
   - Can you recall a moment when your motivation shifted?

4. How confident did you feel using the new skills with actual patients or situations after the training? Did your confidence change over time?

   Probe:

   - What contributed to your confidence level?

5. How relevant or useful did the training feel to your day-to-day responsibilities? Can you share an example of something that felt especially relevant or not relevant?

   Probe:

   - What felt especially applicable to your work?

Training Design:

6. Thinking specifically about the structure and content of the training itself, aside from your motivation or work environment, can you describe a situation where something from the training helped you perform a task differently at work?

   Probes:

   - What part of the training influenced that task?

- How did it change what you did?

7. How closely did the training activities, examples, or scenarios reflect the tasks you actually perform in your job? Can you give an example where similarity or lack of similarity affected your ability to apply the training?

   Probes:

   - What's one example of similarity or mismatch?
   - How did that affect your ability to apply the skill?

8. Which specific elements of the training design (for example, practice activities, examples, feedback, or how the material was organized) most helped you apply what you learned? Were there aspects of the training design that made it harder to apply what you learned, even if your workplace supported the training?
   Probes:
   - What feature was most helpful?
   - Was anything confusing or hard to use?

9. What features of the training most influenced whether you continued using what you learned after the training ended? Did anything about the way the training was presented or structured make it easier to remember or continue using the skills?

   Probe:

   - What made the material easier to remember?

Work Environment:

10. Thinking about your day-to-day work, how does your workplace help or hinder your ability to use what you learned in recent training? Can you share specific examples?
    Probes:

    - Can you share a specific example?
    - What workplace factor mattered most?

11. In what ways does your supervisor influence whether you use new skills or ideas from training in your clinical practice? Can you describe a time when their actions either encouraged or discouraged you from applying what you learned?

    Probes*:*
    - What supervisor action made a difference?
    - How did their response affect your use of the training?

12. How do factors such as your workload, time pressures, and available resources—like staffing levels, organizational policies and procedures, equipment, and documentation or other systems—affect your ability to put training into practice with patients?

    Probe:

    - What resource or constraint impacted you most?

13. How do your colleagues and the broader workplace culture respond when you try to use something new from training? What kinds of reactions or norms make it easier or harder for you to keep using those new approaches?
    Probes:

    - What reaction from colleagues stands out?
    - What cultural norm helped or made it harder?

Closing Question:

14. Is there anything else you'd like to add, or anything else that was not asked, but you wish to share?

Closing Script

Thank you again for taking the time to share your experiences. Your insights are extremely valuable to our project and will help us better understand how allied health professionals apply training in real clinical settings. If you would like a brief summary of our findings once

the project is complete, feel free to let me know. Please also feel free to add anything else you think would be useful for us to understand.